

CE 310 — Week 12 Assignment: Sensitivity Analysis

Due: Wed Nov 18, 9:00 AM — submit to D2L

Overview

The in-class exercise ranked four inputs on one excavation item and found that unit price dominated — not because cost responds more steeply to it, but because it is the input we know least well. This assignment asks whether that holds on a different item, and then scales the question to the level where the money actually is.

Problem 1 is common to everyone. Problem 2 splits by major. Problem 3 is a short written reflection.

Datasets: CE310_BidItems_2024_2026.csv and CE310_BidProjects_2024_2026.csv

How to Submit

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Major declaration	Set MAJOR = "CE" or MAJOR = "ArcE" in the Setup cell — this determines which Problem 2 track is graded
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Written responses	Put each in its own markdown cell immediately after the code cell it responds to
File name	CE310_W12_Assignment_<NetID>.ipynb e.g. CE310_W12_Assignment_abc123.ipynb
Submission	Upload the .ipynb file to D2L → Assignments by 9:00 AM, Wed Nov 18

Problem 1 — A Different Item (30 pts)

Do not repeat the exercise computations. The exercise worked roadway excavation. This problem works **CEM STABIL BKFL** — cement-stabilised backfill, priced per cubic yard, at a design quantity of **500 CY**.

P1a (8 pts) — Fit and Compare the Elasticity

Fit the same model form the exercise used, $\log p \sim \log q + L0W$, to the backfill data. Report the elasticity and the model's residual RMSE in $\log_{10}$ units.

ANSWER_P1a_elasticity = ____ # *logq coefficient, 4 decimals*
ANSWER_P1a_rmse = ____ # *sqrt(mse_resid), 4 decimals*

Written (4 pts). In 3–4 sentences, compare this elasticity to the exercise's -0.1660 for excavation. Which item shows the stronger economy of scale, and offer an engineering reason why — think about what each item's cost is actually driven by. Then say what the RMSE difference implies about which item's price a model can predict more reliably.

P1b (8 pts) — The Tornado on Backfill

Build the same four-input tornado the exercise built, at $Q = 500$ CY, with the same ranges: quantity $\pm 15\%$, unit price ± 1 RMSE, overhead 8–18%, escalation 2–8%.

ANSWER_P1b_base_cost = ____ # *base item cost, nearest dollar*
ANSWER_P1b_swing_price = ____ # *unit-price swing, nearest dollar*

Written (4 pts). Report all four swings. Does the ranking match the exercise's? State the price swing as a percentage of base cost for both items and explain why the two percentages differ — your answer should refer to the RMSE values from P1a, not to the dollar amounts.

P1c (7 pts, written) — Why the Ranking Is Stable

In the exercise, moving from a 2,000 CY job to a 50,000 CY job left the ranking of the four inputs unchanged. Here, moving to an entirely different line item also leaves it unchanged.

In 4–6 sentences, explain **why** — what property of the cost model $Q \times P \times (1+0H) \times (1+Esc)$ makes the ranking insensitive to both the design point and the item? Then describe one specific, realistic change to the inputs that **would** reorder it, and say which two inputs would swap.

P1d (7 pts) — The Survey That Would Be Worth Buying

An estimator can commission a detailed quantity survey that narrows the take-off range from $\pm 15\%$ to $\pm 4\%$. Compute the backfill quantity swing before and after, and the reduction as a percentage of base cost.

ANSWER_P1d_swing_after = ____ # quantity swing at $\pm 4\%$, nearest dollar

Written (4 pts). Report the reduction in dollars. Then answer the question the estimator actually asked: is the survey worth commissioning? Your answer must compare the reduction against the **unit-price** swing that the survey does nothing about, and state what you would need to know about the survey's cost to give a defensible yes or no.

Problem 2 — Track Application (18 pts)

Choose one track. CE students complete Track A. ArcE students complete Track B. Declare your major in the Setup cell; only the matching track is graded.

Track A — CE: Where the Programme's Money Actually Is (18 pts)

One item on one job is not how an agency budgets. A district lets hundreds of contracts a year, and the question is not "which input dominates this job" but "where would a fixed improvement budget do the most good across the programme."

Use the **awarded** excavation work — winning bids only, as in exercise A4.

(A2-a, 6 pts) Compute the programme quantity and programme cost, then the exposure if every unit price were 1 RMSE high.

ANSWER_A2a_prog_cost = ____ # programme cost, nearest dollar

(A2-b, 6 pts) Now compute the programme exposure to take-off error at $\pm 15\%$, and again if a survey programme halved it to $\pm 7.5\%$.

ANSWER_A2b_qty_exposure = ____ # programme quantity exposure at $\pm 15\%$, nearest dollar

(A2-c, 6 pts) Compare the two exposures.

ANSWER_A2c_ratio = ____ # price exposure / quantity exposure at $\pm 15\%$, 2 decimals

In 5–6 sentences: report both exposures and the ratio. A director proposes spending the entire improvement budget on quantity surveying. Using your numbers, say what that would and would not buy. Then identify what would

have to change for surveying to become the right investment, and name one reason the price exposure may be less reducible than the quantity exposure — think about what the RMSE actually represents.

Track B — ArcE: How Long Will It Take, and What Does Late Cost? (18 pts)

Why this track is yours. Contract time is administered by whoever holds the owner's interest during construction, and on building work that is routinely the architect or architectural engineer — through schedule updates, time-extension requests, and the liquidated-damages exposure that hangs on them. The data is highway work because agencies publish it, but the question is the one you will meet on any project with a completion date in the contract.

Use `CE310_BidProjects_2024_2026.csv`, restricted to **Construction** contracts.

(B2-a, 6 pts) Fit a duration model, $\log_{10}(\text{WorkingDays}) \sim \log_{10}(\text{WinningBid_USD})$, and report the slope and R^2 .

ANSWER_B2a_slope = ____ *# slope, 4 decimals*

(B2-b, 6 pts) Predict the contract duration for a **\$2,500,000** contract, then again if the contract value comes in 25% high and 25% low.

ANSWER_B2b_days = ____ *# predicted working days at \$2.5M, nearest day*

(B2-c, 6 pts) Liquidated damages on this contract are **\$12,000 per day**. Compute the cost of the duration swing you found in B2-b — the difference in predicted days between the high and low cases, priced at the LD rate.

ANSWER_B2c_ld_cost = ____ *# LD cost of the duration swing, nearest dollar*

In 5–6 sentences: report the slope, the three durations and the LD figure. The slope is well below 1.0 — say what that means for how duration scales with contract size, and why that is what you would expect on real projects. Then address the limitation that matters most: your model's R^2 is around 0.52, so roughly half the variation in duration is unexplained. State what that does to the confidence you would attach to the LD figure, and what you would want before putting it in a schedule risk register.

Problem 3 — Written Reflection (12 pts)

Write a **100–150 word** reflection addressing the following prompt:

The lecture opened on a slope where nobody had measured ground-water, and the whole decision turned on that one unmeasured input.

Across the exercise and this assignment you have now ranked inputs three times, and unit price won every time for the same reason — not because the answer responds most strongly to it, but because it is what we know least well. Describe **one situation** in construction where the input that dominates a sensitivity ranking is the one you should spend money to pin down, and **one situation** where the dominant input is not worth chasing and you should manage the risk instead. Support each with a **specific number** from this week.

Grading criteria (12 pts total): - First situation: specific, plausible, grounded in a number from this week (4 pts) - Second situation: specific, and explains why more measurement would not help (4 pts) - Writing quality: concise, within 100–150 words, evidence used appropriately (4 pts)

Grading Summary

Problem	Description	Points
P1a	Backfill elasticity and RMSE, compared to excavation	8
P1b	Four-input tornado on backfill; ranking and percentages	8
P1c	Why the ranking is stable, and what would reorder it	7
P1d	The survey that narrows take-off error; is it worth buying	7
P2	Track A (programme exposure) or Track B (duration and LD)	18
P3	Reflection, 100–150 words, two situations with numbers	12
Total		60

Submission Checklist

- ☐ Setup cell run with MAJOR declared
- ☐ P1a: elasticity and RMSE printed + written comparison to excavation
- ☐ P1b: base cost and price swing printed + all four swings reported
- ☐ P1c: written explanation of why the ranking is stable
- ☐ P1d: swing after survey printed + written recommendation
- ☐ P2 (only your declared track): all ANSWER_ values printed and written responses
- ☐ P3: reflection, 100–150 words, two situations, two specific numbers